

BajaSplice

Four sequence models of human splicing, each reported next to the control that decides whether its score means anything. Two of the four clear their controls decisively. **One does not, and that failure is the more useful result.**

Genome GRCh38, GENCODE v50 Splits chromosome-disjoint, test chr1/3/5/7/9
Data 135 GB streamed, 19 GB retained

PREMISE

A number without its control is not a result

Splicing data punishes optimism. Internal exons are 88 percent constitutive, so a model that predicts inclusion for everything scores well. Real splice sites carry GT and AG, so a model that finds dinucleotides looks like it found splice sites. Bound eCLIP windows are GC-rich, so a classifier of GC content looks like a classifier of protein binding. Every one of those is a published-looking number produced by nothing.

So each task here is reported next to a baseline built to reproduce the artifact and nothing else: a motif-only position weight matrix, exon length and GC, the proximal-site bias, an exon's responsiveness to every other perturbation, dinucleotide-matched decoys. The tables below always put the control immediately above the model.

0.935 splice-site PR-AUC, against 0.066 for the motif	0.965 preferred-exon AUC, against 0.749
0.991 ClinVar pathogenic splice variants flagged	0/475 RBPs where the response model beat its reference

INPUTS

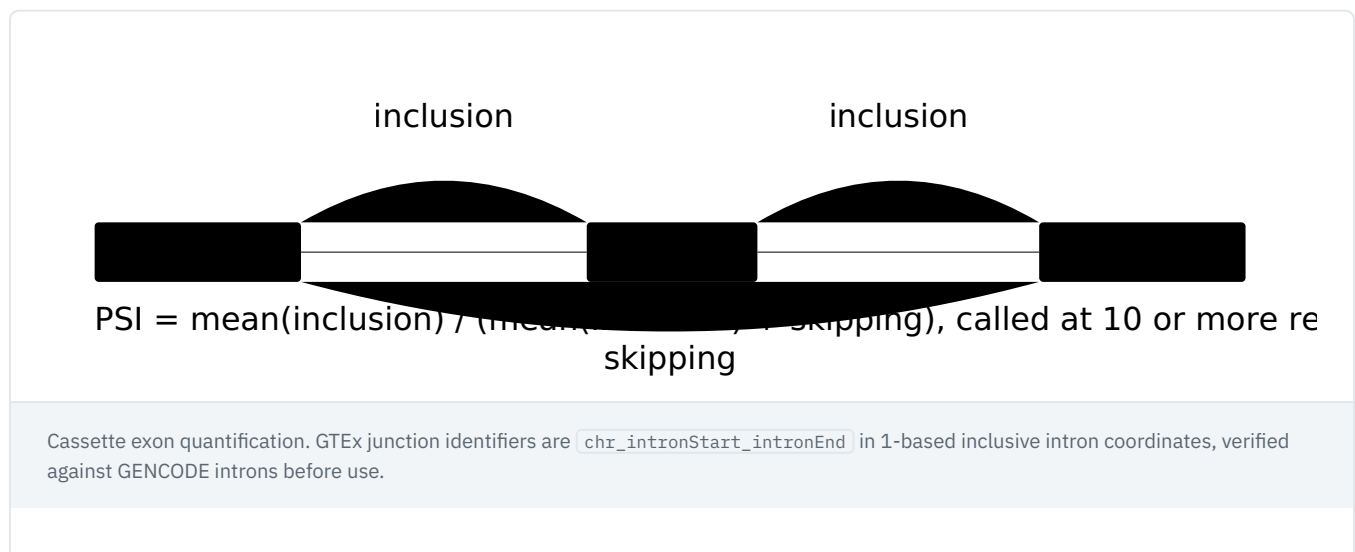
Data

Labels come from RNA-seq rather than annotation. Annotation records which exons exist; it does not record how often each is used, and for cryptic exons it deliberately records nothing at all.

SOURCE	CONTENTS	STREAMED	KEPT
GTEx v8 junctions	357,746 junctions, 17,382 samples, 54 tissues	4.1 GB	4.1 GB
ENCODE RBP knockdowns	983 experiments, 479 RBP targets, K562 and HepG2	118 GB	3.1 GB
recount3 TCGA	junction counts, 33 tumour types	11 GB	11 GB
recount3 TDP-43	3 knockdown studies, junction level	57 MB	57 MB
VastDB hg38	curated PSI, independent quantification	1.2 GB	1.2 GB

ENCODE ships kallisto TSVs of 206,791 transcripts in identical order, so each 30 MB file reduces to one float32 row. Streaming and reducing during download fits the whole collection into 2.6 percent of its footprint.

Inclusion levels are computed from pooled junction reads per tissue. For a cassette exon with flanking constitutive exons C1 and C2, inclusion is the mean of the two flanking junctions and skipping is the junction that bypasses the exon.



This produced 663,089 cassette events across 54 tissues, and 226,235 competing splice sites in 93,598 groups.

RESULT 1

Splice sites

A dilated residual CNN over 2,000 nt of context predicts donor, acceptor or neither at every position of a pre-mRNA. Evaluated across all 397,770,000 positions of the held-out chromosomes, containing 56,075 true acceptors and 56,077 true donors.

MODEL	ACCEPTOR TOP-K	ACCEPTOR PR-AUC	DONOR TOP-K	DONOR PR-AUC
PWM baseline, motif only	0.133	0.066	0.179	0.103
SpliceNet ctx-2000	0.888	0.935	0.885	0.931

The PWM recovers the canonical consensus correctly (CAG|GTAAGT for donors, a polypyrimidine tract before AG| for acceptors) and still reaches only 0.066 PR-AUC, because GT and AG occur millions of times genome-wide. The 14-fold gap is the context the network adds.

RESULT 2

Preferred exons

Given the four splice-site windows of a cassette event and its geometry, predict inclusion in each of 54 tissues. 196,967 held-out events.

MODEL	PEARSON	PREFERRED AUC	ALT-SUBSET R	PER-TISSUE R	MAE
Constant, training mean	n/a	0.500	n/a	n/a	0.145
Geometry only, length / GC / frame	0.211	0.670	0.201	0.217	0.141
Splice-site PWM only	0.192	0.684	0.155	0.196	0.139
Geometry + PWM	0.304	0.749	0.253	0.312	0.135
PSINet ctx-2000	0.764	0.965	0.697	0.779	0.062

Alt-subset r is computed only on the 25,792 test exons with real skipping evidence. Because 87.9 percent of internal exons are constitutive, an overall correlation mostly measures constitutive-versus-not, and the alt-subset column is the one that cannot be inflated by the majority class.

The labels themselves were checked against VastDB, which quantifies the same exons in a different sample panel with a different method.

COMPARISON	EXONS	PEARSON	MAE
All matched exons	141,700	0.908	0.037
Alternative subset	25,136	0.935	0.066

Independent agreement at $r = 0.935$ on genuinely alternative exons indicates the labels reflect biology rather than an artifact of how junctions were counted here.

RESULT 3

Competing splice sites

When several splice sites compete for the same partner, which one wins. Sites are scored independently and a softmax over each group predicts how usage divides. 26,345 held-out groups.

MODEL	TOP-1 ACCURACY	USAGE R	USAGE MAE
Random, by group size	0.450	n/a	n/a
Always the distal site	0.425	n/a	n/a
Always the proximal site	0.471	n/a	n/a
Strongest PWM site	0.512	0.226	0.359
AltSSNet	0.766	0.709	0.213

The proximal-site control matters here because splice-site choice has a well-known proximal bias. It reaches 0.471 against 0.450 for random, so that bias is nearly absent among annotated competing sites, and the model is not exploiting it.

RESULT 4, NEGATIVE

RBP knockdown response

ENCODE knocked down 479 RBPs in two cell lines. Computing inclusion from the transcript quantifications gives a matrix of which exons move when which RBP is removed: 23,497 exons by 477 RBPs, 10.1 million tested pairs, 206,646 responses at 2.04 percent positive rate.

Two checks establish that the matrix is real before any model touches it. Splitting the replicates and recomputing gives $r = 0.646$ over called hits with 86.7 percent sign agreement. Annotated splicing factors rank far above other RBPs (Mann-Whitney $p = 1.7 \times 10^{-12}$), and 19 of the top 50 are splicing factors against 5.8 expected by chance, with U2AF2, SF3B3, SNRNP200, SF1, U2AF1, HNRNPC and SF3B1 at the head of the list.

The matrix is sound. The model built on it is not.

PREDICTOR	MEAN PER-RBP AUC	RBPS	BEATS REFERENCE
Geometry only	0.512	475	0
Sequence CNN	0.573	475	0
Sequence CNN + predicted binding	0.577	475	0
Exon responsiveness, leave-one-RBP-out	0.851	475	reference

The reference is not a sequence baseline. It reads the other 476 knockdown experiments on the same exon, which the models never see. It is here to show that per-RBP AUC is dominated by exon-level responsiveness rather than RBP identity. Restricting to the 50 strongest-effect RBPs does not help: 0.561 against 0.782, still 0 of 50.

Sequence around an exon does not predict which RBP regulates it. The like-for-like comparison, CNN at 0.573 against geometry at 0.512, shows weak sequence signal for whether an exon is generally fragile, not for which protein acts on it. The gap to 0.851 is far too large to be a tuning artifact, though the model was not extensively tuned and validation peaked at epoch 3.

Binding does not predict regulation

The obvious repair is to stop asking sequence to identify the protein and supply predicted binding instead, from an eCLIP model covering 170 RBPs, all of which overlap the knockdown set. Asked directly, with no training: does predicted binding of RBP *k* near an exon identify the exons that respond when *k* is knocked down?

BINDING TRACK USED	MEAN AUC
Matched, the RBP's own binding	0.5491
Mismatched, a different RBP's binding	0.5486
Matched, after removing the shared per-exon component	0.5066

Scoring one RBP's response with another RBP's binding track works equally well. Once the shared component is removed the matched signal is chance ($p = 0.15$). The whole of the 0.549 comes from a general bindability axis, none from RBP identity.

That could mean the binding model lacks RBP specificity, so the same control was run on the binding model's own held-out eCLIP data: 3,050 windows on disjoint chromosomes, 52 RBPs with enough positives, scoring each RBP's labels with other RBPs' output channels.

DESIGN	CHECKPOINT	MATCHED	MISMATCHED	GC ONLY
A. bound vs background	lowfdr	0.892	0.748	0.821
A. bound vs background	enhanced.final	0.866	0.806	0.821
B. bound vs other RBPs' sites	lowfdr	0.698	0.516	0.516
B. bound vs other RBPs' sites	enhanced.final	0.598	0.536	0.516

Design B is the honest test: every window is a real binding site, so bindability is held constant and only RBP identity can help. Both checkpoints pass it, so the specificity is real.

Three things follow. The binding model does distinguish RBPs, decisively for lowfdr at 0.698 against 0.516 mismatched ($p = 1.8 \times 10^{-8}$, 44 of 52 RBPs). Much of the headline number is not specificity: positives are GC 0.535 against 0.410 for background, and GC content alone reaches 0.821 on design A, so the published 0.892 buys only 0.07 over a single scalar. And the two shipped checkpoints differ threefold in specificity, with `enhanced.final` at a 0.061 gap against 0.182 for `lowfdr`, despite being the default.

Repeating the splicing analysis with the more specific checkpoint changes nothing: matched 0.5383 against mismatched 0.5375, and chance after removing the shared component. A binding model that provably distinguishes RBPs still carries no information about which exons respond to removing them. The failure is biological, not instrumental. Binding is not regulation, and knockdown effects are largely indirect.

Cryptic exons

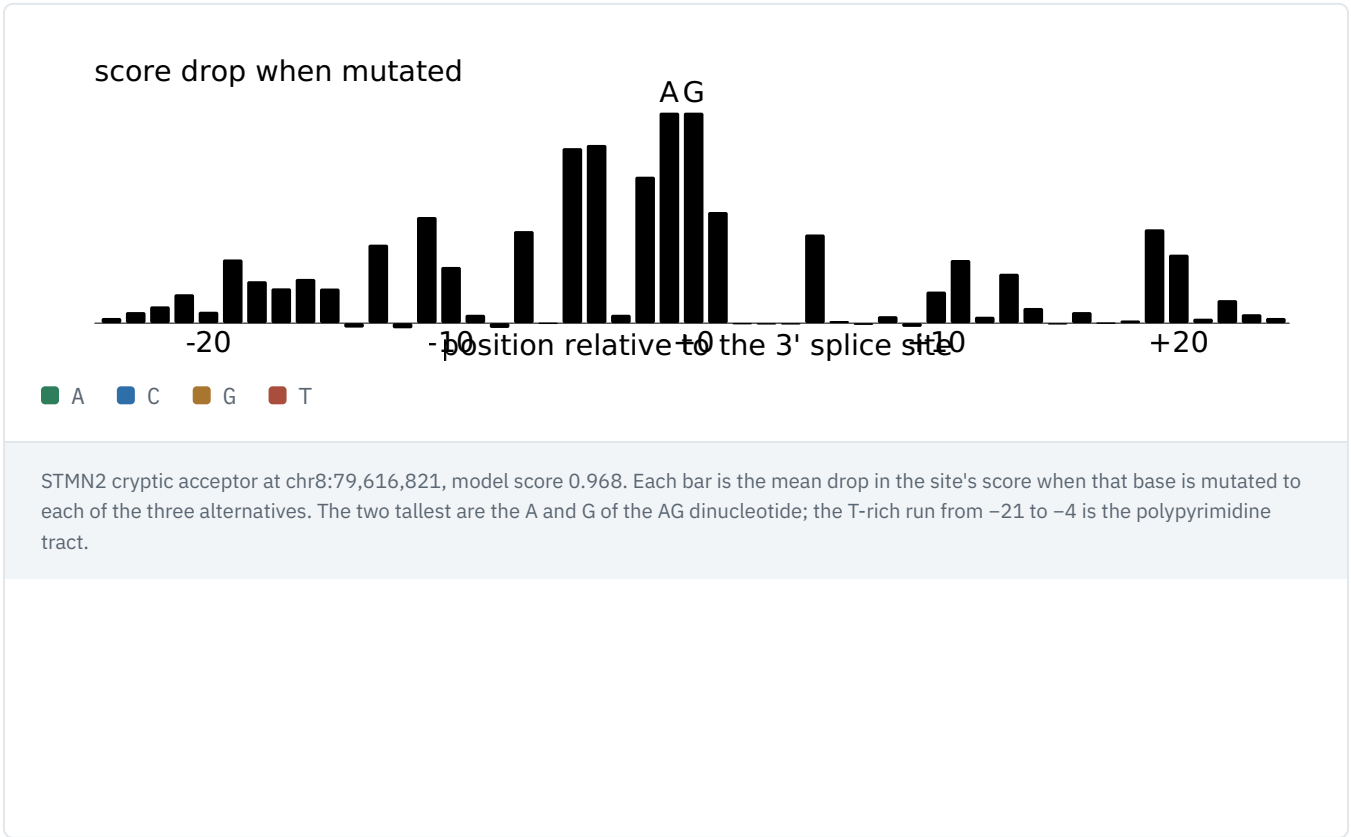
TDP-43 represses cryptic exons, and their appearance is a hallmark of ALS and FTD. They are unannotated by definition, so the inclusion model cannot see them. The splice-site model can, because it scores every position de novo.

Ground truth was built from junction-level RNA-seq rather than annotation. Three TDP-43 knockdown studies were pulled from recount3, whose junction files flag whether each splice site is annotated. Junctions carrying an unannotated site, absent in controls and reproducible after depletion, give 1,883 cryptic junctions and 691 unannotated splice sites on held-out chromosomes.

EVALUATION	SITES	AUC VS MATCHED DECOYS
SRP069787, weak depletion	38	0.632
SRP169127, STMN2 control fires at 88.9×	84	0.838
SRP496259, STMN2 not induced	569	0.885
All held-out sites	691	0.866

Decoys are 13,816 AG or GT dinucleotides drawn from the same introns, so the model gets no credit for the fact that splice sites carry those dinucleotides. 55 percent of cryptic sites outscore every one of their own decoys. These sites sit in training windows labelled negative, so the model was taught to suppress them and still ranks them above matched decoys.

Attribution recovers the mechanism. In-silico mutagenesis on the STMN2 cryptic acceptor, the exon 2a event that produces truncated stathmin-2, puts the largest values on the AG dinucleotide and the polypyrimidine tract that precedes it.



Two genes have published cryptic exon coordinates. STMN2 is recovered decisively, ranking 6th of 3,621 candidate acceptors in its gene at a score of 0.968, against a median of 0.82 for that gene's annotated acceptors. Its near-zero donor score is correct rather than a miss, because exon 2a is a terminal exon ending in a polyadenylation site. UNC13A, whose cryptic exon coordinates come from Ma and colleagues, is enriched but not called: its donor ranks 44th of 4,115 and its acceptors 73rd and 303rd of 6,032, at scores near 0.02.

Rank is usable, probability is not. Only 26 percent of confirmed cryptic sites exceed a score of 0.5, against 0.69 percent of decoys. Screening a gene means ranking its intronic AG and GT dinucleotides and taking the top, not thresholding.

Whether rank tracks how strongly a site is actually used depends on which data is trusted. Pooled across studies there is no relationship (Spearman -0.054 , $p = 0.16$). Within SRP169127, the one study whose STMN2 positive control fires, it is positive but modest ($+0.257$, $p = 0.018$, $n = 84$). Usage after depletion reflects how hard TDP-43 was repressing a site, which is a property of binding rather than of intrinsic splice-site strength.

External validation

Held-out chromosomes guard against memorising the training set. They do not guard against a label pipeline that is wrong in a consistent way, because the test labels came from the same pipeline. The checks below compare model output against measurements made by other people, with each threshold written down before the test was run.

CHECK	EXTERNAL SOURCE	RESULT	REQUIRED	VERDICT
Splice disruption by variant	ClinVar clinical submissions	0.9912	0.90	pass
Novel splice sites in use	GTEx junctions absent from GENCODE	0.8888	0.70	pass
Inclusion levels	VastDB, independent samples and pipeline	0.9348	0.85	pass
TDP-43 cryptic sites	recount3, three depletion studies	0.8659	0.75	pass
Published cryptic exons	Ma et al. 2022; GENCODE v29 vs v50	1.19%	≤ 5%	pass
RBP specificity of binding model	ENCODE eCLIP, held-out windows	0.1820	0.05	pass
Annotated site recovery	GENCODE v50, held-out chromosomes	0.8851	0.85	pass

Clinical pathogenicity ClinVar clinical submissions 0.7178 n/a informational
 Run with `bajasplce validate`. Seven scored checks, all passing; the eighth is reported rather than scored, for the reason below.

Two results carry more than their row. Against ClinVar, 96 percent of pathogenic splice-site variants are flagged as disruptive, against 2.2 percent of benign missense variants elsewhere. And the model recognises splice sites that 17,382 GTEx samples use but GENCODE does not annotate, at 0.889 against dinucleotide-matched decoys, with a median score of 0.915 for real novel sites against 0.000 for decoys.

Those novel GTEx sites score a median of 0.915 while the TDP-43 cryptic sites score 0.009. Both are unannotated, and the difference is biological rather than technical: a junction used in healthy tissue is a strong site the annotation missed, whereas a cryptic exon is an intrinsically weak site that something was holding shut.

What the failing tests taught

Three of these checks failed on their first construction. In each case the test was wrong rather than the model, which is the usual way round and the reason to look before adjusting anything.

i Two questions wearing one label

Comparing pathogenic against benign variants that ClinVar both annotates as splice variants gives 0.72. Both classes sit at splice sites, so both disrupt splicing; what separates them is whether losing that site causes disease in that gene, and the benign ones are in BRCA1, MSH2 and MSH6, where it certainly would. Against benign missense variants the same model scores 0.9912. The test now has two arms: splicing disruption is scored, clinical pathogenicity is reported. The 0.72 is the right answer to a question the model does not answer, and should not be tuned away.

ii A test the data could not answer

Asking whether high-scoring unannotated positions are spliced in GTEx returned zero hits in both the high and the low group. The GTEx v8 junction file is filtered to 357,746 junctions, of which only 3,684 are absent from GENCODE, so the candidates were simply not in it. Inverting the test, using GTEx's own novel junctions as positives against matched decoys, asks the same question of data that can answer it.

iii A pass bought by dropping a case

The published cryptic exon check passed while quietly discarding STMN2, because GENCODE v50 now annotates that exon in a minor isoform and the site was therefore excluded from the candidate list. The check now retries against the canonical-only annotation, the scope a cryptic screen actually runs in, and fails if any published site cannot be scored at all. STMN2 ranks 2nd of 3,664 candidates that way.

Five traps

Each of these produced a confident, wrong answer before it was caught. They are listed because they are properties of this kind of data rather than of this project.

01 The null that was larger than the effect

ENCODE reuses one control across roughly 8.6 knockdowns, so pairs of different controls differ by batch rather than by noise. A control-versus-control null had larger inclusion changes than the knockdowns themselves at every percentile. Taken at face value it says no RBP changes anything. The usable null is each exon's own spread across all knockdowns.

02 The control that read its own answer

Exon responsiveness computed from the full truth matrix includes the RBP being scored. Leave-one-RBP-out moves it from 0.868 to 0.851. Small, but it is the control the model is judged against, so it has to be clean.

03 The reversal that scrambled the track

Scanning a minus-strand gene chunk by chunk and reversing the whole track once at the end looks equivalent to reversing each chunk and is not, because it also reverses chunk order. The symptom was annotated splice sites in UNC13A scoring zero. Any minus-strand gene longer than one chunk was affected.

04 The known sites that filled the ranking

Annotated splice sites sit just outside the exons they belong to, so masking exon bodies does not remove them. They score near 1.0 and occupied the entire top of every cryptic candidate list until excluded explicitly.

05 The classifier that was measuring GC

eCLIP positives are GC 0.535 against 0.410 for background windows. GC content alone separates them at AUROC 0.821, so a reported 0.892 is worth 0.07 over one scalar. Only the positives-versus-other-positives design isolates what the model knows about protein identity.

CAVEATS

Limitations

- The RBP response checkpoints reproduce a negative result. They are not usable predictors.
- The correlation between predicted rank and measured cryptic usage rests on 84 sites from one study, the only one whose positive control fired. Pooled across studies it disappears.
- Only two genes have published cryptic exon coordinates, so the site-level benchmark rests on de novo discovery whose thresholds were relaxed to reach usable numbers.
- Of three TDP-43 knockdown studies, the largest showed no STMN2 induction. A construct overexpressed alongside endogenous TDP-43 is not a clean loss of function, and it supplies 569 of the 691 benchmark sites.
- The inclusion model peaked at epoch 6 of 10 and drifted afterwards. The saved checkpoint is the best one, but it would train better with earlier stopping.
- Both cryptic exon genes sit on training chromosomes. The 691-site benchmark is held out; the two literature cases are not.
- TCGA junction data for 33 tumour types was downloaded and verified but is not used by any model here.

IMPLEMENTATION

Software

`bajasplice` is a Python package of 42 modules. All five trained checkpoints ship inside it, together with a gene index covering 77,081 genes, so the only external requirement is a genome FASTA.

COMPONENT	CONTENTS	SIZE
Model weights	5 checkpoints, 0.36M to 0.82M parameters each	13 MB
Gene index	77,081 genes, canonical transcript, packed exon blobs	10 MB
Source	5,000 lines, 28 tests	0.4 MB

The networks are small because dilated CNNs suffice: the 0.935 PR-AUC splice-site model is 0.36M parameters in a 1.4 MB file. The gene index stores exon coordinates as int32 offsets from the gene start rather than one row each, which is what takes 322 MB down to 10 MB.

Scanning a gene, ranking cryptic candidates and per-base attribution all run from the command line or as a library, and emit track payloads for a genome browser client. The tests pin the coordinate conventions and the chunk-mapping arithmetic, since those are the failures that are silent rather than loud.

Splits follow the SpliceAI convention so figures stay comparable with published work. Inclusion labels derive from GTEx v8 and are validated against VastDB. Knockdown data is ENCODE. Cryptic exon discovery uses recount3 junction counts from SRP169127, SRP069787 and SRP496259. UNC13A cryptic exon coordinates are from Ma et al., Nature 2022; the STMN2 exon 2a coordinates were recovered by differencing GENCODE v29 and v50 rather than taken from a citation.

All figures in this report are measured on held-out chromosomes unless stated otherwise.